A Practical Guide to Bias variance tradeoff

TOPIC -- BIAS VARIANCE TRADEOFF

$$\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Variance} + \sigma_\varepsilon^2$$

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E [(f (x) – E [f ^ (x)]) 2] = E [f (x) 2] – 2 E [f (x) E [f ^ (x)]] + E [E [f ^ (x)] 2] = f (x) 2 – 2 f (x) E [f ^ (x)] + E [f ^ (x)] 2 = (f (x) – E [f ^ (x)]) 2 {\displaystyle {\begin{aligned}{\color {Blue}\mathbb {E} {\Big [}{\big (}f(x)-\mathbb {E} {\big [}{\hat {f}}(x){\big]}{\big)}^{2}{\Big]}}&=\mathbb {E} {\big [}f(x)^{2}{\big]}\,-\,2\ \mathbb {E} {\Big [}f(x)\ \mathbb {E} {\big [}{\hat {f}}(x){\big]}{\Big]}\,+\,\mathbb {E} {\Big [}\mathbb {E} {\big [}{\hat {f}}(x){\big]}^{2}{\Big]}\\&=f(x)^{2}\,-\,2\ f(x)\ \mathbb {E} {\big [}{\hat {f}}(x){\big]}\,+\,\mathbb {E} {\big [}{\hat {f}}(x){\big]}^{2}\\&={\Big (}f(x)-\mathbb {E} {\big [}{\hat {f}}(x){\big]}{\Big)}^{2}\end{aligned}}}

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Approaches Dimensionality reduction and feature selection can decrease variance by simplifying models. Similarly, a larger training set tends to decrease variance. Adding features (predictors) tends to decrease bias, at the expense of introducing additional variance. Learning algorithms typically have some tunable parameters that control bias and variance; for example,

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where N 1 (x) , … , N k (x) {\displaystyle N_{1}(x),\dots ,N_{k}(x)} are the k nearest neighbors of x in the training set. The bias (first term) is a monotone rising function of k, while the variance (second term) drops off as k is increased. In fact, under "reasonable assumptions" the bias of the first-nearest neighbor (1-NN) estimator vanishes entirely as the size of the training set approaches infinity.

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In statistics and machine learning, the bias–variance tradeoff describes the relationship between a model's complexity, the accuracy of its predictions, and how well it can make predictions on previously unseen data that were not used to train the model. In general, as the number of tunable parameters in a model increases, it becomes more flexible, and can better fit a training data set. That is, the model has lower error or lower bias. However, for more flexible models, there will tend to be greater variance to the model fit each time we take a set of samples to create a new training data set. It is said that there is greater variance in the model's estimated parameters. The bias–variance dilemma or bias–variance problem is the conflict in trying to simultaneously minimize these two sources of error that prevent supervised learning algorithms from generalizing beyond their training set:

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MSE = E x ■ [MSE (x)] = E x ■ { Bias D [f ^ (x ; D)] 2 + Var D ■ [f ^ (x ; D)] } + σ 2 . {\displaystyle {\text{MSE}}=\operatorname {\mathbb {E} } _{x}\left[{\text{MSE}}(x)\right]=\operatorname {\mathbb {E} } _{x}\left\{\operatorname {Bias} _{D}\!\left[{\hat {f}}\!(x;D)\right]^{2}+\operatorname {Var} _{D}\left[{\hat {f}}\!(x;D)\right]\right\}+\sigma ^{2}.}

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Derivation The derivation of the bias–variance decomposition for squared error proceeds as follows. For convenience, we drop the D {\displaystyle D} subscript in the following lines, such that f ^ (x ; D) = f ^ (x) {\displaystyle {\hat {f}}\!(x;D)={\hat {f}}\!(x)} . Let us write the mean-squared error of our model:

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In classification The bias–variance decomposition was originally formulated for least-squares regression. For the case of classification under the 0-1 loss (misclassification rate), it is possible to find a similar decomposition, with the caveat that the variance term becomes dependent on the target label. Alternatively, if the classification problem can be phrased as probabilistic classification, then the expected cross-entropy can instead be decomposed to give bias and variance terms with the same semantics but taking a different form. It has been argued that as training data increases, the variance of learned models will tend to decrease, and hence that as training data quantity increases, error is minimised by methods that learn models with lesser bias, and that conversely, for smaller training data quantities it is ever more important to minimise variance.

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The bias–variance tradeoff is a central problem in supervised learning. Ideally, one wants to choose a model that both accurately captures the regularities in its training data, but also generalizes well to unseen data. Unfortunately, it is typically impossible to do both simultaneously. High-variance learning methods may be able to represent their training set well but are at risk of overfitting to noisy or unrepresentative training data. In contrast, algorithms with high bias typically produce simpler models that may fail to capture important regularities (i.e. underfit) in the data. It is an often made fallacy to assume that complex models must have high variance. High variance models are "complex" in some sense, but the reverse need not be true. In addition, one has to be careful how to define complexity. In particular, the number of parameters used to describe the model is a poor measure of complexity. The model f a , b (x) = a sin ■ (b x) {\displaystyle f_{a,b}(x)=a\sin(bx)} has only two parameters (a , b {\displaystyle a,b}) but it can interpolate any number of points by oscillating with a high enough frequency, resulting in both a high bias and high variance. An analogy can be made to the relationship between accuracy and precision. Accuracy is one way of quantifying bias and can intuitively be improved by selecting from only local information. Consequently, a sample will appear accurate (i.e. have low bias) under the aforementioned selection conditions, but may result in underfitting. In other words, test data may not agree as closely with training data, which would indicate imprecision and therefore inflated variance. A graphical example would be a straight line fit to data exhibiting quadratic behavior overall. Precision is a description of variance and generally can only be improved by selecting information from a comparatively larger space. The option to select many data points over a broad sample space is the ideal condition for any analysis. However, intrinsic constraints (whether physical, theoretical, computational, etc.) will always play a limiting role. The limiting case where only a finite number of data points are selected over a broad sample space may result in improved precision and lower variance overall, but may also result in an overreliance on the training data (overfitting). This means that test data would also not agree as closely with the training data, but in this case the reason is inaccuracy or high bias. To borrow from the previous example, the graphical representation would appear as a high-order polynomial fit to the same data exhibiting quadratic behavior. Note that error in each case is measured the same way, but the reason ascribed to the error is different depending on the balance between bias and variance. To mitigate how much information is used from neighboring observations, a model can be smoothed via explicit regularization, such as shrinkage.